Supplementary material

Certified threshold decisions from error-bounded river water-quality records over one-way satellite IoT links

Supplementary Table S1: sites of the sealed confirmation cohort

The 31 sites were selected by metadata only (seed 20261107; the full selection log is results/confirm3/selection.csv)
The days are four-parameter days with a record plan in the confirmation run (10,095 in total).

Supplementary Table S1: Sites of the sealed 2024 cohort by region.

Site	Name	State	Days
Gulf and lower Mississippi			
07048600	White River Near Fayetteville, Ar	AR	144
07075270	South Fork Of Little Red River Near Scotland, Ar	AR	343
07263296	Maumelle River Near Wye, Ar	AR	351
072632966	Lake Maumelle At State Hwy 10 Near Wye, Ar	AR	350
Mid-Atlantic and Appalachia			
01481500	Brandywine Creek At Wilmington, De	DE	345
01638500	Potomac River At Point Of Rocks, Md	MD	288
01645704	Difficult Run Above Fox Lake Near Fairfax, Va	VA	339
01649190	Paint Branch Near College Park, Md	MD	305
01650800	Sligo Creek Near Takoma Park, Md	MD	350
01668000	Rappahannock River Near Fredericksburg, Va	VA	327
03056000	Tygart Valley R At Tygart Dam Nr Grafton, Wv	WV	302
03108500	Ohio River At Montgomery Lock & Dam, Lower Pool	PA	291
03179000	Bluestone River Near Pipestem, Wv	WV	353
Ohio valley and Great Lakes			
03293500	M Fk Beargrass Cr At Lexington Rd At Louisville,Ky	KY	309
03297800	Cedar Creek At Hwy 1442 Near Shepherdsville, Ky	KY	326
03298135	Chenoweth Run At Ruckriegal Parkway, Ky	KY	339
03298250	Cedar Creek At Thixton Road Near Louisville, Ky	KY	289
03321500	Green River At Lock 1 At Spottsville, Ky	KY	314
03353200	Eagle Creek At Zionsville, In	IN	317
04193500	Maumee River At Waterville Oh	OH	250
05568705	Mason Tazewell Drainage D At Cr2350E Nr Topeka, Il	IL	297
411402080492400	Mahoning River Near U.S. Route 422 At Warren Oh	OH	356
Pacific coast			
11044000	Santa Margarita R Nr Temecula Ca	CA	355
11461500	Ef Russian R Nr Calpella Ca	CA	339
11507501	Link River Below Keno Canal, Near Klamath Falls,Or	OR	361
14158850	Mckenzie R Blw Trail Br Dam Nr Belknap Springs, Or	OR	358
14163000	Gate Creek At Vida, Or	OR	356
14183020	North Santiam River Blw Stout Creek, Nr Mehama, Or	OR	359
14202650	Wapato Creek At Sw Gaston Road, At Gaston, Or	OR	357
14206950	Fanno Creek At Durham, Or	OR	360
420853121505501	Klamath River At Miller Island Boat Ramp-Bottom	OR	365

Supplementary Table S2: how far each related-work source was read

Access levels used in the manuscript: read in part means the abstract, introduction, method, and text searches were opened; first page means that only the first page was read. No source was read in full. The search log records the 15 searches (all on 2026-11-03, a general web search limited to United States results) and what did not load.

Supplementary Table S2: Reading level of the sources of Table 1.

Source	Level
Albanese et al. (1996)	Introduction, Definitions 3.1 and 3.2, Theorems 3.3 and 3.5, start of Section IV; proofs not read
Boucheron and Salamatian (2000)	First page (abstract and definition)
Brito et al. (2017), Lin et al. (2019)	Introduction, query model, text searched for minimum, maximum, and threshold
Taurone et al. (2024)	Abstract, method, predicate-filtering section
Cheng et al. (2003)	Abstract and introduction; text searched
Diaz-Cacho et al. (2017)	Abstract and introduction
Muñiz-Cuza et al. (2025), Sun et al. (2025)	Abstract and contribution list
Imai and Iri (1988), Deligiannakis et al. (2007)	Read in earlier rounds of the study (level not recorded in the search log)
Lin et al. (2025, Sensors)	Abstract and introduction (the 2019 Energies paper was not opened)

Supplementary note S3: files and reproduction

Everything below is in the project repository; commands run from the Paper3 folder.

- Frozen plan and freeze record: docs/CONFIRM3_PLAN.md and docs/FREEZE3.md (tag confirm3-freeze).
- Confirmation run: python -I src/p3/confirm3.py data/clean/usgs_new3_2024 confirm3 data/clean/usgs_new3_20 outputs in results/confirm3 (verdicts3.md and verdicts3.csv).
- Results write-up: docs/CONFIRM3_RESULTS.md.
- Tables and figures of the manuscript: python -I src/paper/tables5.py, src/paper/region5.py, src/paper/make_figure and src/paper/supplement5.py.
- Checks: python -I src/paper/check_draft.py draft3 and python -I src/paper/check_numbers5.py.